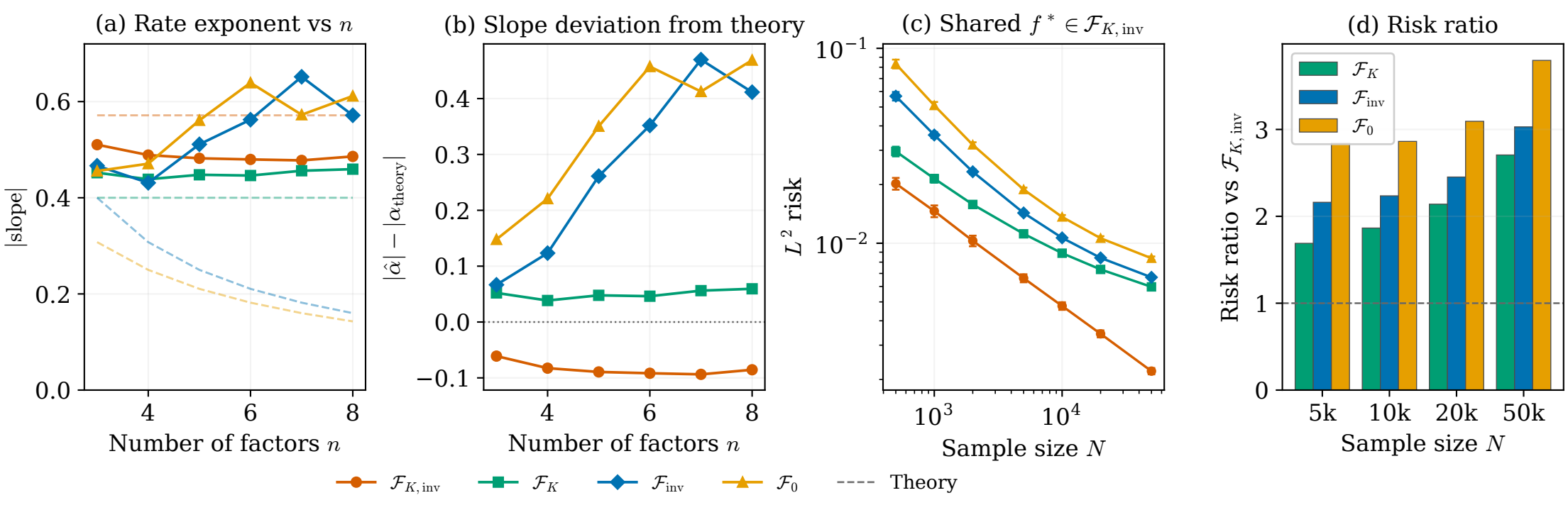
(a) Rate exponent vs n
|slope|
0.0
0.2
0.4
0.6
4
6
8
Number of factors n
(b) Slope deviation from theory
$|\hat{\alpha}| - |\alpha_{\text{theory}}|$
−0.1
0.0
0.1
0.2
0.3
0.4
4
6
8
Number of factors n
(c) Shared $f^* \in \mathcal{F}_{K,\text{inv}}$
L^2 risk
10^{-1}
10^{-2}
10^3
10^4
Sample size N
(d) Risk ratio
Risk ratio vs $\mathcal{F}_{K,\text{inv}}$
0
1
2
3
$\mathcal{F}_K$
$\mathcal{F}_{\text{inv}}$
$\mathcal{F}_0$
5k
10k
20k
50k
Sample size N
$\mathcal{F}_{K,\text{inv}}$
$\mathcal{F}_K$
$\mathcal{F}_{\text{inv}}$
$\mathcal{F}_0$
Theory